

Semester 1 · Academic Year 2026–2027

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Exercise 3. Measuring instruments [1.A · 1.B]

(3 marks)

A voltmeter has its V terminal on B and its COM terminal on A . It displays $+3.0\text{ V}$. Therefore:

- ☐ $u_{AB} = 3.0\text{ V}$;
☐ $u_{AB} = -3.0\text{ V}$;
☐ $V_A = 3.0\text{ V}$;
☐ the voltmeter is wrongly connected, so nothing can be concluded.

To measure the current through a resistor R , an ammeter of internal resistance R_A is connected:

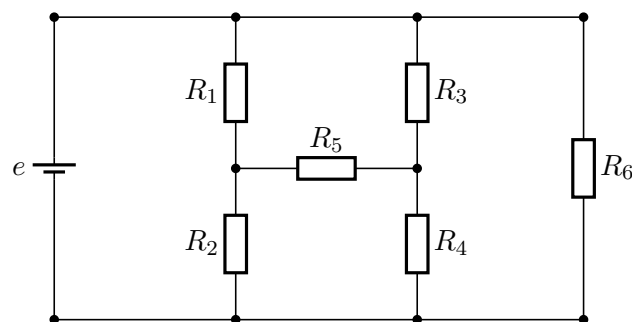
- ☐ in parallel across R , with $R_A \gg R$ ☐ in series with R , with $R_A \ll R$
☐ in series with R , with $R_A \gg R$ ☐ in parallel across R , with $R_A \ll R$

A voltmeter of internal resistance R_V is connected across a resistor R . The measurement is not distorted provided that:

- ☐ $R_V \ll R$ ☐ $R_V = R$ ☐ $R_V \gg R$ ☐ $R_V = 0$

Exercise 4. Reading a circuit diagram [1.C · 3.A]

(4 marks)



Base your answers on the diagram exactly as drawn.

How many nodes?

- ☐ 2 ☐ 4 ☐ 6 ☐ 10

How many branches?

- ☐ 5 ☐ 6 ☐ 7 ☐ 9

How many independent node equations can be written?

- ☐ 3 ☐ 4 ☐ 6 ☐ 7

How many independent meshes?

- ☐ 3 ☐ 4 ☐ 5 ☐ 7

Exercise 5. Characteristics and Ohm's law [2.A · 2.B]

(3 marks)

The characteristic $u = g(i)$ of an ohmic conductor is a straight line through the origin, of slope 2.0 V mA^{-1} . Its conductance is:

- ☐ 2.0 mS ☐ 0.50 mS ☐ 2.0 kS ☐ 500 S

In the generator convention, Ohm's law for a resistor reads:

- ☐ $u = Ri$ ☐ $u = -Ri$ ☐ $u = i/R$ ☐ $u = -i/R$

In the (u, i) plane, a vertical line $u = E$ is the characteristic of:

- ☐ an ideal voltage source ☐ an ideal current source
☐ an ideal diode ☐ an ohmic conductor

Exercise 6. Drift velocity and resistance of a cable [1.B · 2.B · 2.C]

(7 marks)

A copper cable of length L and cross-sectional area S carries a steady current i .

(a) Determine the drift speed v of the conduction electrons. (b) How long does an electron take to travel the whole length of the cable? Why does a lamp at the end of the cable nevertheless light up almost instantly? (c) Determine the resistance R of the cable, then the power p it dissipates.

$L = 10 \text{ m}$	$S = 2.5 \text{ mm}^2$	$i = 4.0 \text{ A}$
$n = 8.0 \times 10^{28} \text{ m}^{-3}$	$e = 1.6 \times 10^{-19} \text{ C}$	$\rho = 1.7 \times 10^{-8} \Omega \text{ m}$

**Marking**

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Exercise 7. Least-squares estimate of a resistance [2.A · 2.B]

(4 marks)

The characteristic of a resistor was measured in the receiver convention:

i_k (mA)	1.0	2.0	3.0	4.0
u_k (V)	2.1	4.1	5.9	8.0

Compute the least-squares estimate $\hat{R} = \sum_k u_k i_k / \sum_k i_k^2$, then the conductance G . What power does the resistor receive at the last measurement point?

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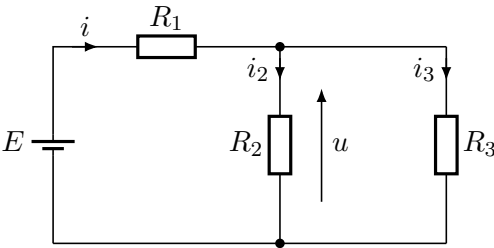
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Exercise 8. A resistive circuit [3.A]

(6 marks)

Determine the equivalent resistance R_{eq} seen by the source, the current i it delivers, the voltage u across R_2 (use a voltage divider), then the currents i_2 and i_3 . Check the node law.



$E = 12\text{ V}$	$R_1 = 2.0\text{ k}\Omega$	$R_2 = 6.0\text{ k}\Omega$	$R_3 = 3.0\text{ k}\Omega$
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